Supplementary Material

Enhanced Double Deep Q-Network (E-DDQN) with Asymmetric Reward Design Versus Standard Deep Q-Network for Safety-Prioritized Cybersickness Detection in Virtual Reality Under an Adversarial Threat Model: A Comparative Analysis

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Revision

Contents

S1 Data, provenance and preprocessing **1**
S1.1 Provenance . . . 1
S1.2 Label . . . 2
S1.3 Linkage groups . . . 2
S1.4 Causal features . . . 2
S1.5 Removed variables . . . 2

S2 Frozen protocol, search spaces and selected configurations **2**
S2.1 Search spaces . . . 2
S2.2 Selected configuration per outer fold . . . 3

S3 Additional results **4**
S3.1 Secondary operating rules . . . 4
S3.2 Group-bootstrap confidence intervals . . . 5
S3.3 Per-fold results . . . 5
S3.4 Event-level behaviour . . . 5
S3.5 Paired tests under the secondary rules . . . 5
S3.6 Value-estimation diagnostics . . . 6
S3.7 Confusion matrices . . . 6

S4 Telemetry-perturbation stress tests **6**
S4.1 Design (frozen before evaluation; `configs/attacks.json`) . . . 6
S4.2 Interpretation limits . . . 6
S4.3 All variants . . . 6

S5 Audit of the submitted version **9**
S5.1 What could and could not be reproduced . . . 9
S5.2 Leakage reproduction . . . 9
S5.3 Code defects (submitted script) . . . 9
S5.4 Internal inconsistencies of the submitted manuscript . . . 9
S5.5 References . . . 10

S6 Reproducibility **10**

S1 Data, provenance and preprocessing

S1.1 Provenance

The evaluation data are the 35 sessions of the UFFCSData curated release (`DATABASE_SELECTED_DATA`, December 2019, seated posture; 9,390 samples at ≈1 Hz). Each curated file was matched sample by sample to one raw export (`RAW_DATABASE`); `dataset.csv`, the file used by the submitted manuscript, is exactly these sessions concatenated in lexicographic folder order (1, 10, 11, . . .), with integer-coded categories and no session identifier. The 37 October-2019 standing-posture raw sessions were used only for code development and budget calibration; their camera-orientation channels are constant within every session (a logging defect). Two December-2019 raw

sessions that the dataset authors excluded were not used. Full per-session table: `results/preprocessing/session_table.csv`.

S1.2 Label

$y_t = 1$ iff the self-reported discomfort level (0–3) at second t is ≥ 1. Prevalence 44.2 % (Flight 41.2 %, Race 48.1 %); 9 of 35 sessions never report discomfort; the median onset is 82 s after the start. Labels are close to step functions within a session (mean 1.1 switches per session), so consecutive samples are strongly dependent.

S1.3 Linkage groups

No participant identifier exists. Sessions are linked when their nine-field questionnaire profile is identical or when one restarts within 180 s after an aborted session (< 200 samples) with at most two differing fields; this gives 21 groups (log: `results/preprocessing/linkage_log.csv`). Groups are the unit of every split.

S1.4 Causal features

Definitions of all 39 telemetry features and 14 questionnaire indicators: `results/preprocessing/feature_dictionary.csv`. Learned parameters per outer fold (imputation value, spike threshold, scaler centre and scale) were fitted on the development groups only (`learned_parameters_by_outer_fold.csv`). Executable checks: truncation invariance of all features (max. difference 0.0 over 40 random cut points); bit-identical preprocessing parameters when test rows are tampered with; zero session, group, identical-profile and duplicated-feature-row overlap between every development set and its test fold (`partition_overlap_checks.csv`).

S1.5 Removed variables

Field of view, static-frame and auto-camera flags (constant within each game: duplicates of the game indicator); haptic feedback, degree of control, DOF simulation, locomotion, posture and symptoms (constant overall); absolute player position (encodes course distance and therefore elapsed time; absent in the October wave); VRSQ scores (post-session).

S2 Frozen protocol, search spaces and selected configurations

The protocol (`configs/protocol.json`) and the perturbation configuration (`configs/attacks.json`) were committed to the repository before any model was trained on the evaluation sessions (commit recorded in `configs/FROZEN_PROTOCOL_COMMIT.txt`). Outer-test rows are not even featurised during hyper-parameter search; the final stage scores each outer-test fold once.

S2.1 Search spaces

Family	Hyper-parameter	Candidate values
All neural models	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
All neural models	hidden	[64, 64], [128, 128], [256, 256], [256, 512, 256]
All neural models	dropout	0.0, 0.1, 0.2, 0.3
All neural models	norm	batch, layer
All neural models	lr	0.0003, 0.001, 0.003
All neural models	weight_decay	0.0, 0.0001, 0.001
All neural models	batch_size	256, 512
All agents	eps_decay	0.9, 0.95, 0.97
All agents	warmup	0, 4000
All agents	updates_per_learn	1, 2, 4
E-DDQN	gamma	0.0, 0.5, 0.9, 0.99
E-DDQN	tau	0.005, 0.01, 0.05
E-DDQN	rho	0.4, 0.5, 0.6
E-DDQN	full_info	False, True
E-CB	rho	0.4, 0.5, 0.6
E-CB	full_info	False, True
DQN	gamma	0.0, 0.5, 0.9, 0.99
DQN	hard_every	10, 100, 500
CS-MLP	sampling	uniform, balanced
LogReg	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
LogReg	C	0.0001, 0.001, 0.01, 0.1, 1.0, 10.0
HGB	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
HGB	learning_rate	0.02, 0.05, 0.1

Family	Hyper-parameter	Candidate values
HGB	max_iter	50, 100, 200, 400
HGB	max_leaf_nodes	4, 8, 16, 31
HGB	min_samples_leaf	20, 100, 300
HGB	l2_regularization	0.0, 1.0, 10.0
RF	feature_set	telemetry, telemetry+profile, exposure, exposure+profile
RF	max_depth	3, 5, 8, None
RF	min_samples_leaf	5, 25, 100, 300
RF	max_features	sqrt, 0.5, 1.0
RF	n_estimators	300
All agents	fixed	epochs=40, eval_every=2, chunk=256, learn_every=2, consolidation=5, eps_start=1.0, eps_end=0.01, buffer=50000, grad_clip=1.0

S2.2 Selected configuration per outer fold

Selection criterion: pooled inner out-of-fold 5:1 expected cost at the cost-minimising threshold (ties: PR-AUC).

Model	Outer	OOF cost	Selected configuration
E-DDQN	0	0.424	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=True, rho=0.4, tau=0.05, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.0, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=4
E-DDQN	1	0.452	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=True, rho=0.4, tau=0.05, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.0, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=4
E-DDQN	2	0.484	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=True, rho=0.4, tau=0.05, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.0, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=4
E-DDQN	3	0.490	feature_set=exposure+profile, hidden=[128, 128], gamma=0.99, full_info=False, rho=0.4, tau=0.005, norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.0001, batch_size=512, eps_decay=0.9, warmup=0, updates_per_learn=1
E-DDQN	4	0.483	feature_set=exposure+profile, hidden=[256, 512, 256], gamma=0.9, full_info=False, rho=0.5, tau=0.05, norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.001, batch_size=512, eps_decay=0.95, warmup=4000, updates_per_learn=2
E-CB	0	0.472	feature_set=exposure, hidden=[256, 512, 256], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.3, lr=0.003, weight_decay=0.001, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=4
E-CB	1	0.459	feature_set=exposure, hidden=[256, 512, 256], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.3, lr=0.003, weight_decay=0.001, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=4
E-CB	2	0.518	feature_set=exposure, hidden=[256, 512, 256], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.3, lr=0.003, weight_decay=0.001, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=4
E-CB	3	0.523	feature_set=exposure, hidden=[128, 128], gamma=0.0, full_info=False, rho=0.6, norm=layer, dropout=0.2, lr=0.001, weight_decay=0.001, batch_size=256, eps_decay=0.97, warmup=0, updates_per_learn=2
E-CB	4	0.466	feature_set=exposure+profile, hidden=[64, 64], gamma=0.0, full_info=True, rho=0.5, norm=layer, dropout=0.2, lr=0.003, weight_decay=0.001, batch_size=256, eps_decay=0.95, warmup=0, updates_per_learn=1
DQN	0	0.432	feature_set=exposure, hidden=[64, 64], gamma=0.0, full_info=False, hard_every=10, norm=layer, dropout=0.0, lr=0.001, weight_decay=0.0001, batch_size=256, eps_decay=0.95, warmup=0, updates_per_learn=2
DQN	1	0.452	feature_set=exposure+profile, hidden=[256, 512, 256], gamma=0.9, full_info=False, hard_every=500, norm=layer, dropout=0.0, lr=0.001, weight_decay=0.001, batch_size=512, eps_decay=0.9, warmup=0, updates_per_learn=4
DQN	2	0.538	feature_set=exposure, hidden=[128, 128], gamma=0.9, full_info=False, hard_every=500, norm=batch, dropout=0.3, lr=0.0003, weight_decay=0.0, batch_size=256, eps_decay=0.95, warmup=0, updates_per_learn=2
DQN	3	0.501	feature_set=exposure, hidden=[128, 128], gamma=0.9, full_info=False, hard_every=100, norm=layer, dropout=0.2, lr=0.001, weight_decay=0.0, batch_size=512, eps_decay=0.97, warmup=0, updates_per_learn=2

Model	Outer	OOF cost	Selected configuration
DQN	4	0.469	feature_set=exposure+profile, hidden=[256, 512, 256], gamma=0.0, full_info=False, hard_every=100, norm=batch, dropout=0.1, lr=0.003, weight_decay=0.0, batch_size=512, eps_decay=0.97, warmup=4000, updates_per_learn=1
CS-MLP	0	0.485	feature_set=exposure+profile, hidden=[128, 128], norm=layer, dropout=0.3, lr=0.001, weight_decay=0.0, batch_size=256, sampling=balanced
CS-MLP	1	0.490	feature_set=exposure+profile, hidden=[128, 128], norm=layer, dropout=0.3, lr=0.001, weight_decay=0.0, batch_size=256, sampling=balanced
CS-MLP	2	0.503	feature_set=exposure, hidden=[64, 64], norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.0, batch_size=256, sampling=uniform
CS-MLP	3	0.527	feature_set=telemetry, hidden=[64, 64], norm=layer, dropout=0.0, lr=0.003, weight_decay=0.0001, batch_size=256, sampling=balanced
CS-MLP	4	0.500	feature_set=telemetry, hidden=[64, 64], norm=layer, dropout=0.3, lr=0.0003, weight_decay=0.001, batch_size=512, sampling=uniform
LogReg	0	0.486	feature_set=exposure, C=0.0001
LogReg	1	0.522	feature_set=telemetry+profile, C=1.0
LogReg	2	0.540	feature_set=exposure, C=10.0
LogReg	3	0.511	feature_set=exposure, C=0.0001
LogReg	4	0.531	feature_set=exposure+profile, C=0.01
HGB	0	0.486	feature_set=exposure, learning_rate=0.02, max_iter=50, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=1.0
HGB	1	0.501	feature_set=exposure, learning_rate=0.02, max_iter=50, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=1.0
HGB	2	0.542	feature_set=exposure, learning_rate=0.02, max_iter=400, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=0.0
HGB	3	0.532	feature_set=telemetry, learning_rate=0.05, max_iter=200, max_leaf_nodes=31, min_samples_leaf=100, l2_regularization=0.0
HGB	4	0.527	feature_set=exposure, learning_rate=0.02, max_iter=400, max_leaf_nodes=4, min_samples_leaf=20, l2_regularization=0.0
RF	0	0.465	feature_set=telemetry, min_samples_leaf=100, max_depth=3, max_features=0.5
RF	1	0.478	feature_set=exposure, min_samples_leaf=25, max_depth=8, max_features=sqrt
RF	2	0.509	feature_set=exposure, min_samples_leaf=5, max_depth=5, max_features=sqrt
RF	3	0.514	feature_set=exposure, min_samples_leaf=5, max_depth=5, max_features=sqrt
RF	4	0.510	feature_set=telemetry+profile, min_samples_leaf=25, max_depth=3, max_features=sqrt

S3 Additional results

S3.1 Secondary operating rules

Rule D0 (untuned Bayes cut-off $\hat{p} \geq 1/6$; for the asymmetric agents this is the greedy policy):

Model	Recall	Spec.	Prec.	F1	Bal. acc.	MCC	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC	FN	FP
E-DDQN	77.2±3.8	28.0±2.0	45.9±1.7	57.6±2.4	52.6±2.7	0.060±0.061	0.905±0.092	0.538±0.016	0.483±0.019	909	3,862
E-CB	85.9±2.2	37.2±3.0	52.0±1.1	64.7±1.1	61.5±1.5	0.258±0.031	0.663±0.044	0.653±0.011	0.562±0.006	627	3,262
DQN	85.1±1.7	37.1±1.5	51.7±0.7	64.3±0.9	61.1±1.0	0.248±0.022	0.680±0.035	0.661±0.013	0.595±0.022	561	3,330
CS-MLP	83.0±3.1	36.4±2.3	50.8±1.0	63.0±1.5	59.7±1.6	0.215±0.037	0.731±0.066	0.644±0.018	0.521±0.022	622	3,381
LogReg	85.3±0.0	13.6±0.0	43.8±0.0	57.9±0.0	49.4±0.0	−0.016±0.000	0.808±0.000	0.626±0.000	0.576±0.000	611	4,531
HGB	92.0±0.0	30.0±0.0	51.0±0.0	65.6±0.0	61.0±0.0	0.272±0.000	0.567±0.000	0.692±0.000	0.593±0.000	331	3,670
RF	94.5±0.1	34.9±0.1	53.4±0.0	68.3±0.0	64.7±0.0	0.353±0.001	**0.486±0.001**	0.688±0.002	0.548±0.001	228	3,410
Always alarm	100.0	0.0	44.2	61.3	50.0	0.000	0.558	–	–	0	5,243
Never alarm	0.0	100.0	–	–	50.0	0.000	2.208	–	–	4,147	0

Validation recall constraint ($\geq$0.95 on pooled inner-OOF predictions):

Model	Recall	Spec.	Prec.	F1	Bal. acc.	MCC	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC	FN	FP
E-DDQN	86.8±3.1	11.5±3.4	43.7±0.9	58.1±1.3	49.1±1.7	−0.027±0.053	0.785±0.062	0.538±0.016	0.483±0.019	641	4,539
E-CB	95.2±2.7	15.0±3.3	47.0±0.8	62.9±1.0	55.1±1.4	0.167±0.042	0.581±0.051	0.653±0.011	0.562±0.006	317	4,093
DQN	93.5±2.9	23.0±3.4	49.0±0.8	64.3±1.0	58.3±1.2	0.227±0.032	0.573±0.052	0.661±0.013	0.595±0.022	343	3,810
CS-MLP	98.3±1.2	9.3±4.2	46.2±1.0	62.8±0.9	53.8±1.9	0.156±0.052	0.544±0.025	0.644±0.018	0.521±0.022	105	4,463
LogReg	97.2±0.0	18.3±0.0	48.5±0.0	64.7±0.0	57.7±0.0	0.241±0.000	0.518±0.000	0.626±0.000	0.576±0.000	116	4,285
HGB	97.5±0.0	18.3±0.0	48.6±0.0	64.8±0.0	57.9±0.0	0.248±0.000	0.511±0.000	0.692±0.000	0.593±0.000	103	4,282
RF	97.0±0.2	25.4±0.3	50.7±0.0	66.6±0.0	61.2±0.0	0.307±0.002	**0.483±0.004**	0.688±0.002	0.548±0.001	128	3,896
Always alarm	100.0	0.0	44.2	61.3	50.0	0.000	0.558	–	–	0	5,243
Never alarm	0.0	100.0	–	–	50.0	0.000	2.208	–	–	4,147	0

S3.2 Group-bootstrap confidence intervals

Seed-ensemble predictions, rule D1; 95 % percentile intervals from 2,000 bootstrap replicates of the 21 linkage groups.

Model (seed ensemble, D1)	Recall	Spec.	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC
E-DDQN	90.4 [77.7, 98.9]	8.1 [1.5, 17.6]	0.726 [0.533, 0.996]	0.541 [0.382, 0.690]	0.496 [0.329, 0.683]
E-CB	96.6 [92.2, 99.6]	14.6 [8.3, 21.4]	0.552 [0.431, 0.673]	0.659 [0.550, 0.767]	0.565 [0.429, 0.740]
DQN	96.9 [92.6, 99.7]	16.3 [8.7, 24.8]	0.536 [0.417, 0.654]	0.665 [0.558, 0.785]	0.606 [0.456, 0.778]
CS-MLP	99.4 [98.5, 100.0]	6.1 [2.6, 10.8]	0.537 [0.421, 0.634]	0.654 [0.527, 0.781]	0.529 [0.393, 0.712]
LogReg	99.1 [98.0, 100.0]	8.1 [4.0, 13.0]	0.532 [0.429, 0.627]	0.626 [0.498, 0.750]	0.576 [0.408, 0.754]
HGB	97.0 [92.4, 99.9]	14.6 [7.0, 24.9]	0.543 [0.439, 0.654]	0.692 [0.574, 0.809]	0.593 [0.437, 0.779]
RF	98.7 [97.5, 99.6]	18.3 [11.7, 28.0]	0.485 [0.381, 0.577]	0.688 [0.582, 0.808]	0.548 [0.443, 0.735]

S3.3 Per-fold results

Expected 5:1 cost / ROC-AUC of the seed ensemble in each outer fold (rule D1). Fold-to-fold variation is large because each fold contains only 3–5 linkage groups.

Model	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
E-DDQN	0.604 / 0.536	0.762 / 0.522	0.529 / 0.734	1.184 / 0.501	0.693 / 0.466
E-CB	0.569 / 0.655	0.537 / 0.718	0.462 / 0.756	0.460 / 0.747	0.758 / 0.447
DQN	0.564 / 0.715	0.757 / 0.521	0.451 / 0.852	0.426 / 0.740	0.520 / 0.499
CS-MLP	0.604 / 0.575	0.542 / 0.596	0.495 / 0.820	0.528 / 0.708	0.501 / 0.659
LogReg	0.582 / 0.623	0.542 / 0.428	0.486 / 0.810	0.531 / 0.712	0.515 / 0.610
HGB	0.619 / 0.631	0.547 / 0.740	0.510 / 0.791	0.522 / 0.761	0.495 / 0.585
RF	0.600 / 0.630	0.463 / 0.728	0.397 / 0.771	0.457 / 0.761	0.486 / 0.685

S3.4 Event-level behaviour

Model	Events detected	Median delay (s)	Alarm % before onset	Alarm % in event-free sessions
E-DDQN	26/26	0	82.3	96.6
E-CB	25/26	0	73.1	76.9
DQN	25/26	0	77.4	70.4
CS-MLP	26/26	0	92.9	88.1
LogReg	25/26	0	76.0	91.7
HGB	25/26	0	69.3	90.8
RF	25/26	0	61.3	75.3

S3.5 Paired tests under the secondary rules

Rule D0:

B (vs. A = E-DDQN)	b	c	err_A	err_B	χ^2_{cc}	p_{exact}	$\Delta\text{Cost}_{5:1}$ (A−B) [95% CI]	p_{group}
E-CB	635	1,517	4,771	3,889	360.7	<0.001	+0.214 [+0.025, +0.487]	0.018
DQN	467	1,347	4,771	3,891	425.9	<0.001	+0.242 [+0.039, +0.542]	0.012
CS-MLP	431	1,199	4,771	4,003	360.9	<0.001	+0.204 [−0.030, +0.507]	0.11
LogReg	1,297	926	4,771	5,142	61.6	<0.001	+0.087 [−0.293, +0.541]	0.69
HGB	833	1,603	4,771	4,001	242.8	<0.001	+0.328 [+0.078, +0.639]	0.003
RF	526	1,659	4,771	3,638	586.5	<0.001	+0.411 [+0.147, +0.748]	0.001

Validation recall ≥ 0.95:

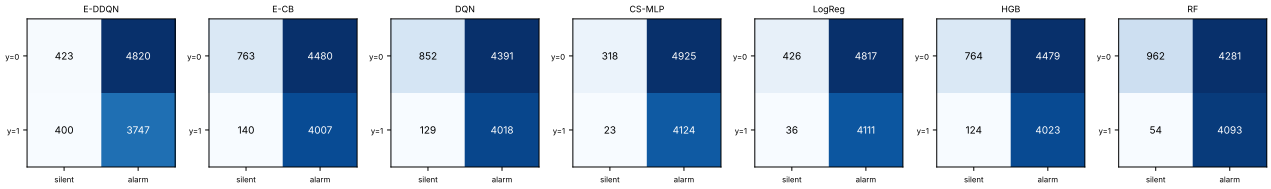
B (vs. A = E-DDQN)	b	c	err_A	err_B	χ^2_{cc}	p_{exact}	$\Delta\text{Cost}_{5:1}$ (A−B) [95% CI]	p_{group}
E-CB	378	1,148	5,180	4,410	387.5	<0.001	+0.220 [+0.017, +0.566]	0.021
DQN	294	1,321	5,180	4,153	651.8	<0.001	+0.236 [+0.013, +0.580]	0.028
CS-MLP	509	1,121	5,180	4,568	229.0	<0.001	+0.294 [+0.033, +0.652]	0.013
LogReg	538	1,317	5,180	4,401	326.3	<0.001	+0.307 [+0.027, +0.687]	0.02
HGB	434	1,229	5,180	4,385	379.1	<0.001	+0.314 [+0.030, +0.711]	0.023
RF	393	1,549	5,180	4,024	686.9	<0.001	+0.342 [+0.067, +0.719]	0.003

S3.6 Value-estimation diagnostics

Mean Q-value of the greedy action on the inner-validation groups compared with the discounted return actually obtained by the greedy policy on those sessions (rewards of every action are known offline).

Model	γ (selected)	mean $Q(s, \pi(s))$	mean MC return	bias (mean ± SD over runs)
DQN	0–0.9	1.467	1.074	$+0.393 \pm 1.493$
E-CB	0–0	0.224	0.136	$+0.088 \pm 0.078$
E-DDQN	0.9–0.99	5.998	8.898	-2.900 ± 8.250

S3.7 Confusion matrices



Seed-ensemble predictions, rule D1, all 9,390 outer-test samples (rows: true label; columns: decision).

S4 Telemetry-perturbation stress tests

S4.1 Design (frozen before evaluation; `configs/attacks.json`)

Perturbations act on the raw per-second channels of the outer-test sessions; causal features are then recomputed with the training-fitted featurizer and scaler of each inner model. Cybersickness labels are never modified; the indicator z marks perturbed seconds. Stress scenarios (each over the whole session or one random 60-s window): frame-rate cap at 72, 60 or 45 fps; uniform orientation jitter ±0.02, 0.05 or 0.10 per quaternion component; speed up-scaling by 10, 25 or 50 % with acceleration recomputed as the per-second speed difference (as in the logger); speed spikes of 10, 20 or 40 units every 20 s. Evasion-only candidates (whole session): speed down-scaling by 25 or 50 %, orientation smoothing with $\lambda = 0.5$ or 0.25, frame-rate restoration to 90 fps. All values are clipped to the per-game ranges observed in the data. Field-of-view manipulation is not evaluated because the field of view never varies in the data.

S4.2 Interpretation limits

The tests measure how bounded manipulation of the logged channels moves the detectors' scores and decisions. They do not show that a manipulation would cause or hide cybersickness in a person, because no physiological effect was measured. The adaptive worst cases give the adversary the labels and choose the most damaging candidate per session and model.

S4.3 All variants

Recall and false-positive rate on perturbed seconds (%) and 5:1 cost on those seconds, clean vs. perturbed, seed ensembles at the D1 operating point.

Model	Scenario	Value	Window	Rec. clean	Rec. att.	FPR clean	FPR att.	Cost clean	Cost att.
CS-MLP	accel spikes	10	whole	99.4	99.6	93.9	95.5	0.537	0.542
CS-MLP	accel spikes	10	window60	99.3	99.7	90.5	92.4	0.532	0.536
CS-MLP	accel spikes	20	whole	99.4	99.8	93.9	98.3	0.537	0.553
CS-MLP	accel spikes	20	window60	100.0	100.0	89.2	97.9	0.545	0.599
CS-MLP	accel spikes	40	whole	99.4	99.9	93.9	99.1	0.537	0.556
CS-MLP	accel spikes	40	window60	100.0	100.0	97.0	99.6	0.496	0.510
CS-MLP	fps cap	45	whole	99.4	100.0	93.9	98.3	0.537	0.549
CS-MLP	fps cap	45	window60	99.5	100.0	95.4	98.8	0.509	0.514
CS-MLP	fps cap	60	whole	99.4	100.0	93.9	98.3	0.537	0.549
CS-MLP	fps cap	60	window60	98.6	100.0	95.4	99.0	0.649	0.649
CS-MLP	fps cap	72	whole	99.4	100.0	93.9	98.3	0.537	0.549
CS-MLP	fps cap	72	window60	99.5	100.0	96.6	98.8	0.511	0.510
CS-MLP	fps restore	90	whole	99.4	99.7	93.9	93.7	0.537	0.530
CS-MLP	rot jitter	0.02	whole	99.4	99.1	93.9	92.0	0.537	0.534
CS-MLP	rot jitter	0.02	window60	99.9	99.7	96.7	94.6	0.517	0.511
CS-MLP	rot jitter	0.05	whole	99.4	98.8	93.9	91.4	0.537	0.536
CS-MLP	rot jitter	0.05	window60	98.3	96.4	94.4	91.2	0.656	0.667
CS-MLP	rot jitter	0.1	whole	99.4	99.7	93.9	96.5	0.537	0.545

Model	Scenario	Value	Window	Rec. clean	Rec. att.	FPR clean	FPR att.	Cost clean	Cost att.
CS-MLP	rot jitter	0.1	window60	100.0	99.8	94.3	96.2	0.560	0.576
CS-MLP	rot smooth	0.25	whole	99.4	99.5	93.9	97.1	0.537	0.553
CS-MLP	rot smooth	0.5	whole	99.4	99.3	93.9	96.8	0.537	0.555
CS-MLP	speed scale down	−0.5	whole	99.4	95.6	93.9	85.3	0.537	0.573
CS-MLP	speed scale down	−0.25	whole	99.4	98.5	93.9	90.2	0.537	0.538
CS-MLP	speed scale up	0.1	whole	99.4	99.6	93.9	95.1	0.537	0.540
CS-MLP	speed scale up	0.1	window60	99.3	99.7	92.2	94.1	0.541	0.544
CS-MLP	speed scale up	0.25	whole	99.4	100.0	93.9	96.7	0.537	0.541
CS-MLP	speed scale up	0.25	window60	100.0	100.0	95.3	97.2	0.534	0.544
CS-MLP	speed scale up	0.5	whole	99.4	100.0	93.9	98.2	0.537	0.548
CS-MLP	speed scale up	0.5	window60	100.0	100.0	87.4	97.1	0.448	0.497
DQN	accel spikes	10	whole	96.9	96.8	83.7	86.1	0.536	0.552
DQN	accel spikes	10	window60	92.6	92.6	80.3	82.7	0.617	0.631
DQN	accel spikes	20	whole	96.9	98.3	83.7	95.5	0.536	0.570
DQN	accel spikes	20	window60	96.9	97.4	78.5	98.0	0.540	0.650
DQN	accel spikes	40	whole	96.9	97.9	83.7	95.4	0.536	0.578
DQN	accel spikes	40	window60	99.5	100.0	87.5	99.8	0.460	0.511
DQN	fps cap	45	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	fps cap	45	window60	94.1	94.1	84.5	84.5	0.581	0.581
DQN	fps cap	60	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	fps cap	60	window60	94.5	94.5	83.7	83.7	0.643	0.643
DQN	fps cap	72	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	fps cap	72	window60	93.5	93.5	83.9	83.9	0.591	0.591
DQN	fps restore	90	whole	96.9	96.9	83.7	83.7	0.536	0.536
DQN	rot jitter	0.02	whole	96.9	94.2	83.7	78.0	0.536	0.563
DQN	rot jitter	0.02	window60	95.6	95.5	85.1	83.4	0.555	0.548
DQN	rot jitter	0.05	whole	96.9	93.2	83.7	74.7	0.536	0.568
DQN	rot jitter	0.05	window60	92.5	78.9	82.7	74.8	0.676	0.852
DQN	rot jitter	0.1	whole	96.9	94.8	83.7	83.5	0.536	0.581
DQN	rot jitter	0.1	window60	96.3	95.7	86.6	84.7	0.590	0.591
DQN	rot smooth	0.25	whole	96.9	99.6	83.7	92.4	0.536	0.525
DQN	rot smooth	0.5	whole	96.9	98.9	83.7	91.0	0.536	0.533
DQN	speed scale down	−0.5	whole	96.9	92.2	83.7	70.6	0.536	0.567
DQN	speed scale down	−0.25	whole	96.9	94.7	83.7	76.9	0.536	0.547
DQN	speed scale up	0.1	whole	96.9	97.0	83.7	88.1	0.536	0.557
DQN	speed scale up	0.1	window60	92.6	92.6	84.6	86.5	0.640	0.651
DQN	speed scale up	0.25	whole	96.9	97.4	83.7	90.8	0.536	0.564
DQN	speed scale up	0.25	window60	97.2	97.0	82.6	87.9	0.523	0.558
DQN	speed scale up	0.5	whole	96.9	97.8	83.7	91.8	0.536	0.562
DQN	speed scale up	0.5	window60	99.5	99.5	76.4	86.4	0.403	0.455
E-CB	accel spikes	10	whole	96.6	97.9	85.4	88.3	0.552	0.540
E-CB	accel spikes	10	window60	97.0	98.2	77.4	79.1	0.506	0.491
E-CB	accel spikes	20	whole	96.6	98.9	85.4	97.0	0.552	0.566
E-CB	accel spikes	20	window60	95.0	96.0	76.4	94.8	0.564	0.657
E-CB	accel spikes	40	whole	96.6	99.3	85.4	97.7	0.552	0.562
E-CB	accel spikes	40	window60	99.5	100.0	89.4	99.3	0.469	0.508
E-CB	fps cap	45	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	fps cap	45	window60	96.0	96.0	84.2	84.2	0.533	0.533
E-CB	fps cap	60	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	fps cap	60	window60	94.6	94.6	87.2	87.2	0.663	0.663
E-CB	fps cap	72	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	fps cap	72	window60	97.0	97.0	86.2	86.2	0.518	0.518
E-CB	fps restore	90	whole	96.6	96.6	85.4	85.4	0.552	0.552
E-CB	rot jitter	0.02	whole	96.6	96.5	85.4	76.3	0.552	0.504
E-CB	rot jitter	0.02	window60	97.9	95.5	85.3	81.9	0.502	0.540
E-CB	rot jitter	0.05	whole	96.6	94.4	85.4	79.1	0.552	0.565
E-CB	rot jitter	0.05	window60	93.9	86.8	84.6	71.4	0.664	0.695
E-CB	rot jitter	0.1	whole	96.6	96.1	85.4	88.2	0.552	0.578
E-CB	rot jitter	0.1	window60	97.7	95.0	88.5	82.7	0.572	0.593
E-CB	rot smooth	0.25	whole	96.6	95.9	85.4	94.4	0.552	0.618
E-CB	rot smooth	0.5	whole	96.6	95.3	85.4	90.9	0.552	0.612
E-CB	speed scale down	−0.5	whole	96.6	81.0	85.4	59.5	0.552	0.752
E-CB	speed scale down	−0.25	whole	96.6	88.9	85.4	71.2	0.552	0.643
E-CB	speed scale up	0.1	whole	96.6	98.0	85.4	91.0	0.552	0.552
E-CB	speed scale up	0.1	window60	97.4	98.2	79.4	82.2	0.509	0.508
E-CB	speed scale up	0.25	whole	96.6	98.2	85.4	94.1	0.552	0.565
E-CB	speed scale up	0.25	window60	95.6	95.4	84.8	91.4	0.572	0.614
E-CB	speed scale up	0.5	whole	96.6	98.7	85.4	95.9	0.552	0.564
E-CB	speed scale up	0.5	window60	99.1	99.5	73.3	89.3	0.397	0.469
E-DDQN	accel spikes	10	whole	90.4	90.5	91.9	92.1	0.726	0.723
E-DDQN	accel spikes	10	window60	91.7	91.4	97.9	98.0	0.737	0.745
E-DDQN	accel spikes	20	whole	90.4	94.9	91.9	93.8	0.726	0.637
E-DDQN	accel spikes	20	window60	88.8	90.4	94.5	96.0	0.796	0.774
E-DDQN	accel spikes	40	whole	90.4	95.4	91.9	94.3	0.726	0.627
E-DDQN	accel spikes	40	window60	94.0	94.0	98.7	99.9	0.651	0.657
E-DDQN	fps cap	45	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	fps cap	45	window60	90.2	90.2	99.3	99.3	0.752	0.752
E-DDQN	fps cap	60	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	fps cap	60	window60	83.7	83.7	95.5	95.5	0.907	0.907
E-DDQN	fps cap	72	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	fps cap	72	window60	89.1	89.1	99.2	99.2	0.775	0.775
E-DDQN	fps restore	90	whole	90.4	90.4	91.9	91.9	0.726	0.726
E-DDQN	rot jitter	0.02	whole	90.4	92.7	91.9	91.4	0.726	0.671
E-DDQN	rot jitter	0.02	window60	91.4	92.7	100.0	100.0	0.734	0.702
E-DDQN	rot jitter	0.05	whole	90.4	93.9	91.9	91.2	0.726	0.644
E-DDQN	rot jitter	0.05	window60	83.2	82.1	88.5	88.5	0.870	0.889
E-DDQN	rot jitter	0.1	whole	90.4	94.0	91.9	92.2	0.726	0.648
E-DDQN	rot jitter	0.1	window60	91.0	84.7	88.5	91.4	0.708	0.854
E-DDQN	rot smooth	0.25	whole	90.4	88.0	91.9	91.8	0.726	0.777

Model	Scenario	Value	Window	Rec. clean	Rec. att.	FPR clean	FPR att.	Cost clean	Cost att.
E-DDQN	rot smooth	0.5	whole	90.4	88.9	91.9	92.3	0.726	0.760
E-DDQN	speed scale down	−0.5	whole	90.4	86.7	91.9	90.9	0.726	0.802
E-DDQN	speed scale down	−0.25	whole	90.4	88.2	91.9	91.7	0.726	0.772
E-DDQN	speed scale up	0.1	whole	90.4	91.4	91.9	92.0	0.726	0.704
E-DDQN	speed scale up	0.1	window60	91.3	92.0	98.4	98.4	0.748	0.734
E-DDQN	speed scale up	0.25	whole	90.4	92.8	91.9	91.8	0.726	0.672
E-DDQN	speed scale up	0.25	window60	87.0	88.3	99.8	99.9	0.845	0.817
E-DDQN	speed scale up	0.5	whole	90.4	93.5	91.9	90.7	0.726	0.650
E-DDQN	speed scale up	0.5	window60	94.0	94.0	91.0	90.9	0.612	0.611
HGB	accel spikes	10	whole	97.0	97.1	85.4	87.8	0.543	0.555
HGB	accel spikes	10	window60	99.3	99.3	86.1	88.5	0.507	0.521
HGB	accel spikes	20	whole	97.0	97.9	85.4	94.5	0.543	0.575
HGB	accel spikes	20	window60	91.2	92.8	81.8	95.3	0.670	0.724
HGB	accel spikes	40	whole	97.0	98.1	85.4	95.7	0.543	0.577
HGB	accel spikes	40	window60	99.5	100.0	90.1	97.6	0.473	0.500
HGB	fps cap	45	whole	97.0	97.0	85.4	85.9	0.543	0.545
HGB	fps cap	45	window60	93.9	93.9	89.0	89.2	0.609	0.610
HGB	fps cap	60	whole	97.0	97.0	85.4	85.9	0.543	0.545
HGB	fps cap	60	window60	97.5	97.5	92.3	92.4	0.648	0.649
HGB	fps cap	72	whole	97.0	97.0	85.4	85.9	0.543	0.545
HGB	fps cap	72	window60	99.4	99.4	88.4	88.6	0.471	0.472
HGB	fps restore	90	whole	97.0	97.0	85.4	86.9	0.543	0.551
HGB	rot jitter	0.02	whole	97.0	97.0	85.4	85.0	0.543	0.541
HGB	rot jitter	0.02	window60	99.4	99.4	87.9	87.7	0.482	0.481
HGB	rot jitter	0.05	whole	97.0	96.6	85.4	82.9	0.543	0.538
HGB	rot jitter	0.05	window60	91.6	90.2	84.6	82.0	0.703	0.710
HGB	rot jitter	0.1	whole	97.0	96.3	85.4	82.7	0.543	0.543
HGB	rot jitter	0.1	window60	91.3	89.9	84.0	77.4	0.676	0.664
HGB	rot smooth	0.25	whole	97.0	97.0	85.4	88.3	0.543	0.558
HGB	rot smooth	0.5	whole	97.0	96.9	85.4	88.0	0.543	0.559
HGB	speed scale down	−0.5	whole	97.0	81.0	85.4	64.4	0.543	0.780
HGB	speed scale down	−0.25	whole	97.0	91.0	85.4	78.3	0.543	0.635
HGB	speed scale up	0.1	whole	97.0	97.3	85.4	88.9	0.543	0.556
HGB	speed scale up	0.1	window60	99.4	99.4	81.9	83.3	0.479	0.487
HGB	speed scale up	0.25	whole	97.0	98.0	85.4	90.7	0.543	0.550
HGB	speed scale up	0.25	window60	92.9	94.3	88.1	92.3	0.649	0.643
HGB	speed scale up	0.5	whole	97.0	98.2	85.4	91.7	0.543	0.551
HGB	speed scale up	0.5	window60	99.4	99.4	77.3	87.1	0.411	0.461
LogReg	accel spikes	10	whole	99.1	99.2	91.9	92.6	0.532	0.536
LogReg	accel spikes	10	window60	99.4	99.4	88.9	89.5	0.520	0.524
LogReg	accel spikes	20	whole	99.1	99.3	91.9	94.8	0.532	0.545
LogReg	accel spikes	20	window60	99.4	99.4	86.4	92.6	0.540	0.578
LogReg	accel spikes	40	whole	99.1	99.9	91.9	96.7	0.532	0.543
LogReg	accel spikes	40	window60	99.5	99.5	89.2	93.6	0.468	0.491
LogReg	fps cap	45	whole	99.1	97.0	91.9	86.3	0.532	0.549
LogReg	fps cap	45	window60	99.5	86.8	93.0	82.4	0.496	0.745
LogReg	fps cap	60	whole	99.1	98.1	91.9	88.4	0.532	0.535
LogReg	fps cap	60	window60	99.3	89.7	92.4	84.3	0.618	0.729
LogReg	fps cap	72	whole	99.1	98.6	91.9	90.2	0.532	0.534
LogReg	fps cap	72	window60	99.5	95.2	91.7	88.3	0.486	0.573
LogReg	fps restore	90	whole	99.1	99.1	91.9	91.9	0.532	0.532
LogReg	rot jitter	0.02	whole	99.1	99.2	91.9	92.1	0.532	0.533
LogReg	rot jitter	0.02	window60	99.5	99.5	91.5	91.6	0.499	0.500
LogReg	rot jitter	0.05	whole	99.1	99.3	91.9	92.5	0.532	0.533
LogReg	rot jitter	0.05	window60	96.4	96.8	92.2	92.5	0.673	0.668
LogReg	rot jitter	0.1	whole	99.1	99.2	91.9	92.7	0.532	0.535
LogReg	rot jitter	0.1	window60	95.7	96.9	91.1	91.3	0.629	0.605
LogReg	rot smooth	0.25	whole	99.1	99.1	91.9	91.7	0.532	0.532
LogReg	rot smooth	0.5	whole	99.1	99.1	91.9	91.7	0.532	0.532
LogReg	speed scale down	−0.5	whole	99.1	97.8	91.9	86.8	0.532	0.534
LogReg	speed scale down	−0.25	whole	99.1	98.7	91.9	89.2	0.532	0.526
LogReg	speed scale up	0.1	whole	99.1	99.2	91.9	92.9	0.532	0.536
LogReg	speed scale up	0.1	window60	99.4	99.4	87.8	88.7	0.513	0.518
LogReg	speed scale up	0.25	whole	99.1	99.4	91.9	93.6	0.532	0.536
LogReg	speed scale up	0.25	window60	99.4	99.4	90.8	91.9	0.520	0.527
LogReg	speed scale up	0.5	whole	99.1	99.7	91.9	94.9	0.532	0.537
LogReg	speed scale up	0.5	window60	99.5	99.5	86.0	92.5	0.452	0.486
RF	accel spikes	10	whole	98.7	99.0	81.7	84.5	0.485	0.494
RF	accel spikes	10	window60	99.3	99.3	75.0	79.1	0.444	0.467
RF	accel spikes	20	whole	98.7	99.7	81.7	96.2	0.485	0.543
RF	accel spikes	20	window60	97.6	99.1	68.6	96.2	0.466	0.605
RF	accel spikes	40	whole	98.7	99.8	81.7	97.4	0.485	0.549
RF	accel spikes	40	window60	99.5	100.0	81.1	99.8	0.427	0.511
RF	fps cap	45	whole	98.7	98.7	81.7	81.3	0.485	0.483
RF	fps cap	45	window60	99.4	99.4	84.9	84.9	0.456	0.456
RF	fps cap	60	whole	98.7	98.7	81.7	81.3	0.485	0.483
RF	fps cap	60	window60	99.2	99.2	84.7	84.7	0.569	0.569
RF	fps cap	72	whole	98.7	98.7	81.7	81.3	0.485	0.483
RF	fps cap	72	window60	99.4	99.4	85.6	85.6	0.457	0.457
RF	fps restore	90	whole	98.7	98.8	81.7	81.8	0.485	0.483
RF	rot jitter	0.02	whole	98.7	98.2	81.7	79.4	0.485	0.482
RF	rot jitter	0.02	window60	99.4	99.4	85.0	83.0	0.467	0.456
RF	rot jitter	0.05	whole	98.7	97.7	81.7	79.0	0.485	0.491
RF	rot jitter	0.05	window60	96.1	91.5	84.3	82.4	0.625	0.691
RF	rot jitter	0.1	whole	98.7	98.0	81.7	81.1	0.485	0.496
RF	rot jitter	0.1	window60	96.3	92.8	82.1	79.1	0.563	0.616
RF	rot smooth	0.25	whole	98.7	99.0	81.7	85.9	0.485	0.502
RF	rot smooth	0.5	whole	98.7	99.1	81.7	84.9	0.485	0.494
RF	speed scale down	−0.5	whole	98.7	90.6	81.7	61.8	0.485	0.553

Model	Scenario	Value	Window	Rec. clean	Rec. att.	FPR clean	FPR att.	Cost clean	Cost att.
RF	speed scale down	−0.25	whole	98.7	95.9	81.7	73.3	0.485	0.501
RF	speed scale up	0.1	whole	98.7	99.1	81.7	84.3	0.485	0.492
RF	speed scale up	0.1	window60	99.0	99.0	75.5	77.5	0.452	0.464
RF	speed scale up	0.25	whole	98.7	99.1	81.7	86.7	0.485	0.504
RF	speed scale up	0.25	window60	98.6	98.3	84.0	85.5	0.502	0.515
RF	speed scale up	0.5	whole	98.7	99.0	81.7	92.1	0.485	0.537
RF	speed scale up	0.5	window60	99.4	99.4	66.3	85.6	0.354	0.453

S5 Audit of the submitted version

S5.1 What could and could not be reproduced

The repository contained the raw dataset, the E-DDQN script and the manuscript PDF, but not the 41-feature train/validation/test files, preprocessing code, baseline, ablation, multi-seed or statistics code, trained models or logs. Exact reproduction was therefore impossible. The partition sizes of the submitted paper (14,680 / 3,146 / 3,146; 20,972 in total; 3,670 training samples per grade) are reproduced exactly by oversampling every discomfort grade of the curated release to the size of grade 0 (5,243) and then splitting at row level.

S5.2 Leakage reproduction

Model	Evaluation protocol	Acc.	Recall	Spec.	MCC	ROC-AUC
HGB	Oversample to 5,243/grade, then 70/15/15 row split (submitted protocol)	97.6	99.8	91.0	0.935	0.995
HGB	Row-level random split, no oversampling	95.3	94.1	96.3	0.905	0.992
HGB	Grouped 5-fold split (session-linkage groups)	52.2	33.6	66.9	0.005	0.523
E-DDQN (submitted code)	Submitted script, submitted protocol (proxy)	95.2	99.2	83.1	0.870	–
E-DDQN (submitted code)	Submitted script, grouped split (proxy)	41.1	92.1	7.7	−0.004	–

Under the reconstructed submitted protocol, 2,170 of the 2,359 positive test rows (92 %) have an exact duplicate in the training split and all sessions occur in both splits.

S5.3 Code defects (submitted script)

1. Next state = next row of a shuffled chunk; no terminal flags; $\gamma = 0.99$ bootstrapped across unrelated samples.
2. Teacher forcing stored the true label as the action together with the reward of the random action that was actually taken.
3. The Polyak update ignored batch-normalisation buffers; the target network normalised with mean 0 and variance 1 throughout training.
4. The Double-DQN arg-max ran the online network in training mode.
5. The test set was scored and printed every 10 episodes.
6. Warm-up filled 14,680 transitions (manuscript: 25,000); learning every 25 chunks (manuscript: 50).
7. Comments described 2.5:1 and 1.67:1 rewards and an accuracy-based selection rule that actually maximised F1.
8. SHAP attributed $Q(s, \mathit{Attack})$ (which contains $V(s)$) on 50 test rows with the additivity check disabled.

S5.4 Internal inconsistencies of the submitted manuscript

- McNemar test: with 172 baseline and 98 E-DDQN errors, $b - c$ must equal 74, whereas the reported $b = 96$, $c = 34$ give 62; $\chi^2 = 28.6$ was computed correctly from these impossible counts.
- "Test set evaluated exactly once" versus per-checkpoint test results (Table 6, Figs. 7, 10, 13).
- "The reward is the only change relative to the baseline" versus 14 differences listed in Table 5; baseline trained for 150 episodes with last-episode selection versus 500 episodes with validation selection.
- Baseline described both as a 2-output network and as a native 4-class model.
- MCC and F2 described as threshold-independent.
- "Removing teacher forcing leaves performance essentially unchanged" while the ablation table showed improvements on every metric.

- A 1-s sliding window at 1 Hz contains one sample; an FOV-modulation feature is identically zero in these data.

Arithmetic within Tables 7–9 of the submitted version was internally consistent (`results/audit/manuscript_number_check.csv`).

S5.5 References

Nine references could not be found and fifteen had incorrect bibliographic details or misattributed claims; the numbers attributed to Kundu et al. (93.5 %, ∼35 ms) are not supported by any accessible source. The full verification table is in `reports/06_reference_verification.md`.

S6 Reproducibility

- Software: Python 3.13.16, PyTorch 2.14.1 (CPU), scikit-learn 1.9.1, NumPy 2.5.3, pandas 3.0.5, SciPy 1.18.1, SHAP 0.53.0, statsmodels 0.15.0 (`revision/requirements.txt`).
- One command regenerates everything from the raw XML files: `bash revision/run_all.sh -fresh` (about 10 CPU-hours on four cores, dominated by the reinforcement-learning hyper-parameter search).
- Seeds: split 20261012 (outer) and 20261013–20261017 (inner); hyper-parameter search 777 (+family index for configuration sampling); final training 1000–1004; perturbations 4242; bootstrap 11 (CIs) and 5 (paired tests).
- Every number in the manuscript and in this supplement is produced by `scripts/05_make_tables.py` from `results/tables/*.csv`, which `scripts/04_aggregate.py` computes from the per-sample predictions in `results/predictions/`. `scripts/08_consistency_check.py` recomputes confusion matrices, metrics and McNemar statistics with independent library implementations and checks them against the manuscript macros.
- Executable sanity tests of the corrected agent: `scripts/00_sanity_tests.py` (reward regret ratio, implied threshold, within-session transitions, terminal flags, reward/action consistency of replay, target-network buffer synchronisation, Q-value bounds, McNemar identity).